

“Body Oracle” Translator: Translating Body Postures into Speculative Hieroglyphs

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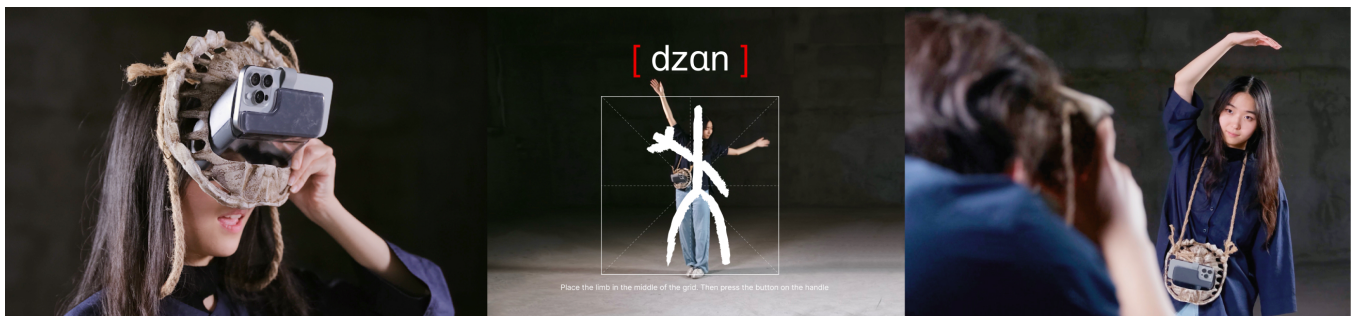


Figure 1: “Body Oracle” Translator is a handheld mixed-reality mask that translates human body postures into speculative hieroglyphs, called “Body Oracle”, in real time. These AI-generated hieroglyphic characters are inspired by Chinese oracle-bone inscriptions and are derived from bodily and inter-bodily spatial positioning.

Abstract

“Body Oracle” Translator is a binocular-like handheld mixed reality mask that translates human body postures into speculative AI-generated hieroglyphic characters called “Body Oracle” in real time. This computational linguistic speculative design project creates an alternative language system inspired by Chinese Oracle-Bone Inscriptions, based on bodily and inter-bodily spatial positioning, promoting cross-cultural collective bodily awareness in an era dominated by language-based communication. “Body Oracle” Translator allows audiences to embody an “alternative present”—one where they believe these hieroglyphic characters have existed for thousands of years—reshaping our shared understanding of language, culture, and community. It challenges viewers to reconsider how embodied communication can enhance our somatic understanding and influence collective body awareness.

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CCS Concepts

• **Human-centered computing** → **Interaction design theory, concepts and paradigms**; **Mixed / augmented reality**.

Keywords

Speculative Design, Chinese Oracle-bone Inscriptions, Somatic Awareness, Cultural Heritage, Alternative Present, Mixed Reality, AI Generation, Somaesthetics, Future Archaeology

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Across both alphabetic (Western) and logographic (Eastern) traditions, most languages lack a precise lexicon for describing complex bodily postures and the body’s spatial movements. Expressing a particular pose typically demands lengthy, inefficient phrasing and remains ambiguous—especially in describing intercorporeal relations and spatial positioning. More broadly, languages lack precise vocabulary for proprioception and social proprioception, our awareness of individual and collective bodily positioning in space [Keysers et al. 2010]. This limitation hinders our ability to articulate

embodied cognition in today’s communication, which remains predominantly text-based and—despite emojis—often “disembodied” [Fuchs 2021], lacking the crucial dimension of bodily awareness.

In response, “Body Oracle” [Huang et al. 2025] is a computational linguistic speculative design that imagines an “alternative present” [Auger 2013] in which a language system for complex proprioceptive experiences exists. Inspired by Chinese oracle-bone inscriptions [Keightley 1985]—one of the oldest forms of Chinese writing—we draw on their elegant pictographic affordances, in which simplified strokes mirror the objects they denote. “Body Oracle” encodes bodily postures and inter-body spatial relations using the aesthetic principles of these ancient inscriptions. The system generates AI-derived hieroglyphic characters representing corporeal poses and intercorporeal configurations. These cross-cultural, intuitive symbols—akin to emojis—aim to expand how we conceptualize, share, and communicate bodily experience.

We present this speculative design in the form of “future archaeology” [Price and Feinman 2001]: an experiential exhibition that traces the imagined evolution of these hieroglyphs and envisions a world where proprioception forms the foundation of linguistic expression—one in which these speculative signs appear to have existed for thousands of years. Within this immersive alternative present, audiences are invited to interact with one another through the “Body Oracle” Translator, broaden their bodily awareness, and imagine a culture transformed by embodied communication.

“Body Oracle” Translator is a handheld optical see-through mixed reality mask that converts human body postures into Body Oracle characters in real time. When a user points the translator at someone, it uses a fine-tuned YOLOv11 pose-estimation model¹ running on an iPhone to recognize body poses. The system matches these poses to characters in the “Body Oracle” Archive (a dictionary index pre-generated by the “Body Oracle” AI system, see Figure 2) and displays the corresponding character in the center of the view. An AI-generated audio of ancient Chinese pronunciation can be triggered with an on-screen button at the top of the iPhone screen. The mask is designed as a binocular, oracle-bone-shaped enclosure for HoloKit [Hu et al. 2024], a smartphone-based, open-source optical see-through stereoscopic headset (see Figure 3). The shell is fabricated from a ready-made oracle-bone form (e.g., a plastic mock-up of a turtle shell). An iPhone is mounted inside with the rear camera unobstructed, allowing environment sensing while preserving optical see-through. A hemp cord allows the device to be worn around the neck when not in use. The Translator immerses participants in an experiential alternative present, encouraging them to interact with one another (see Figure 1) by making different poses in the exhibition.

This experience explores how a somatic language could enhance bodily understanding, foster collective body awareness, revitalize ancient heritage within an experiential alternative present, and point toward somaesthetic futures. Following Wittgenstein’s insight—“*The limits of my language mean the limits of my world*”—we contend that introducing alternative somatic hieroglyphs may reshape collective semantic cognition of bodily awareness. As a sociolinguistic mediator, “Body Oracle” serves as a “true oracle” that

helps us conceptualize, describe, and ultimately transform social reality through a re-embodied approach to language.

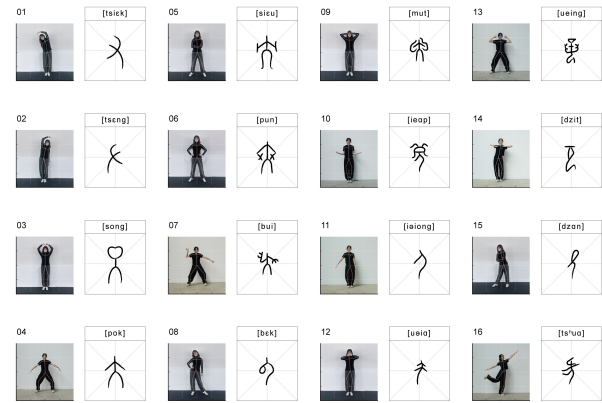


Figure 2: Dictionary Excerpts from the “Body Oracle” Archive



Figure 3: “Body Oracle” Translator device

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¹<https://docs.ultralytics.com/tasks/pose/>